

Right-angled triangles



1

a AB

b BC

2

a AC

b BC

c BC

d AC

e $\angle ACB$

3

a i $\frac{4}{3}$ ii $\frac{4}{5}$ iii $\frac{3}{5}$

b i $\frac{15}{8}$ ii $\frac{15}{17}$ iii $\frac{8}{17}$

c i $\frac{21}{20}$ ii $\frac{21}{29}$ iii $\frac{20}{29}$

d i $\frac{12}{5}$ ii $\frac{12}{13}$ iii $\frac{5}{13}$

e i $\frac{24}{7}$ ii $\frac{24}{25}$ iii $\frac{7}{25}$

f i $\frac{28}{45}$ ii $\frac{28}{53}$ iii $\frac{45}{53}$

Trigonometric ratios

4

a i $\tan \theta = \frac{AC}{BC}$ ii $\sin \alpha = \frac{BC}{AB}$

b i $\sin \theta = \frac{AC}{AB}$ ii $\tan \alpha = \frac{BC}{AC}$

c i $\cos \theta = \frac{BC}{AB}$ ii $\cos \alpha = \frac{AC}{AB}$

d i $\tan \theta = \frac{8}{15}$ ii $\sin \alpha = \frac{15}{17}$

e i $\cos \theta = \frac{15}{17}$ ii $\tan \alpha = \frac{15}{8}$

f i $\sin \theta = \frac{24}{25}$ ii $\cos \alpha = \frac{24}{25}$

5

a $x = 10$

b $\frac{4}{5}$

c $\frac{3}{5}$

6

a $x = 4$

b $\frac{4}{3}$



Unknown sides

7 $b = 8$

8 $d = 3084$

9 $b = 19.8$

10 a $f = 4.08$

b $f = 6.12$

c $g = 9.46$

d $a = 11.59$

Unknown angles

11 $\frac{\sqrt{15}}{7}$

12 a $x = 67^\circ$

b $x = 74^\circ$

13 $A = 66.4^\circ$

14 $\tan \theta = \frac{24}{7}$

Applications

15 a $h = 21$ b $f = 22$ c $h = 98, x = 28$ d $x = 34$

e $x = 12$

16 $d = 1.48$

17 1.96 m

18 $y = 58.02$

19

a $x = 21.85$



b $h = 23.95 \text{ m}$

20 $\theta = 9.88^\circ$

21 $y = 54.90^\circ$

22 $\tan \theta = \frac{4}{3}$

23 $h = 4.4 \text{ m}$